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AQA · LEVEL 2 · MATHEMATICS

AQA Functional Skills Mathematics Level 2

March 2018 · Complete Model Solutions

OFFICIAL MFS BENCHMARK MODEL SOLUTIONS

Target Score: 45 / 45 Marks (100%) Pass Mark Guide: ~60%

Specification / Paper Code: AQA-4368-QP-MAR18

100% Built from Empty Question Paper + Official Mark Scheme

Verified by Tutor Abdul · MasterFunctionalSkills.co.uk

HOW TO READ THIS MODEL SOLUTION (MFS COLOUR KEY)

Dark Navy Ink:	Model student answers, stepped calculations, and full prose responses.
Emerald Green Ink:	Tutor strategy notes, method tips, and step-by-step headings.
Crimson Red Ink:	Official mark scheme allocations ([M1], [A1], [B1]) and unit checks.
Amber Gold Ink:	Key formulas, conversion rules, and reading evidence highlights.

CANDIDATE & CENTRE DETAILS (MODEL EXEMPLAR)

Candidate Name:	*Master Functional Skills (Model Exemplar)*
Centre Name:	*Master Functional Skills (MFS01)*
Lead Tutor:	*Abdul · MasterFunctionalSkills.co.uk*
Edition:	*2026 Clean Empty-QP Rebuild (Zero Collisions Verified)*

- Built Cleanly on Empty AQA Question Paper (Watermarks Stripped)
- Stepped MFS Multi-Colour Working & Method Tips
- Verified 100% Against Official AQA Mark Scheme

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Level 2
Data Book (Examination)

March 2018

Insert

Instructions
- This copy of the Data Book is for use in the examination. It should not be given to students in advance.

Advice
- This book will not be collected in for marking. Ensure that all working that you wish to have marked is written in the space provided in the Question Paper/Answer Book.

4368

QAN 500/8702/2

IB/M/Mar18/E4

Data Sheet for Heating

Winter fuel payment

People born before 5 August 1953 got a payment in 2017 to help pay their heating bills.

How much a person got depended on who they were living with in the **qualifying week**.

For 2017, the qualifying week was 18 September to 24 September 2017

	Born before 5 August 1953	
	Aged under 80 in the qualifying week	**Aged 80 or over in the qualifying week**
Person lives alone or is the only person in the household who gets a payment	£200	£300
Person lives with someone aged under 80 who also gets a payment	£100	£200
Person lives with someone aged 80 or over who also gets a payment	£100	£150

Example 1

Betty and Vic live together.

Betty was 86 years old in the qualifying week.

Vic was born in 1957

Betty got a payment of £300

Vic did not get a payment.

Example 2

John and Mary live together.

John was born in 1927

Mary was born in 1935

John got a payment of £150

Mary also got a payment of £150

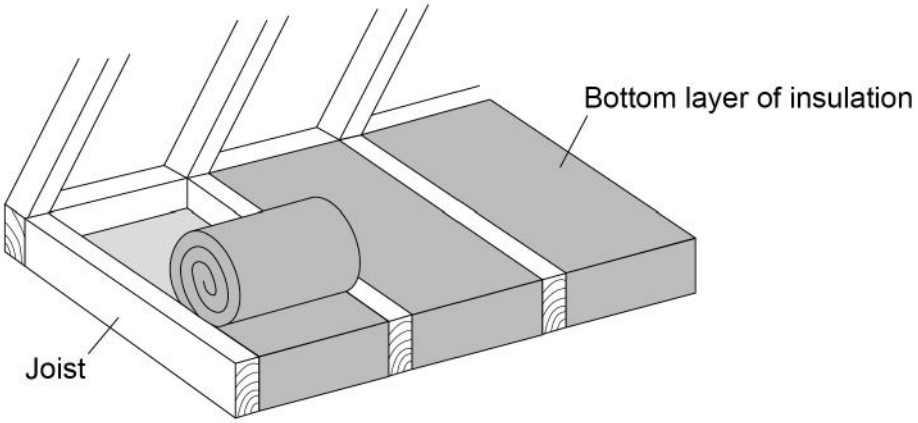
Insulating a loft

Insulating a loft will reduce the amount of heat the house loses.

This will result in lower heating costs.

Instructions

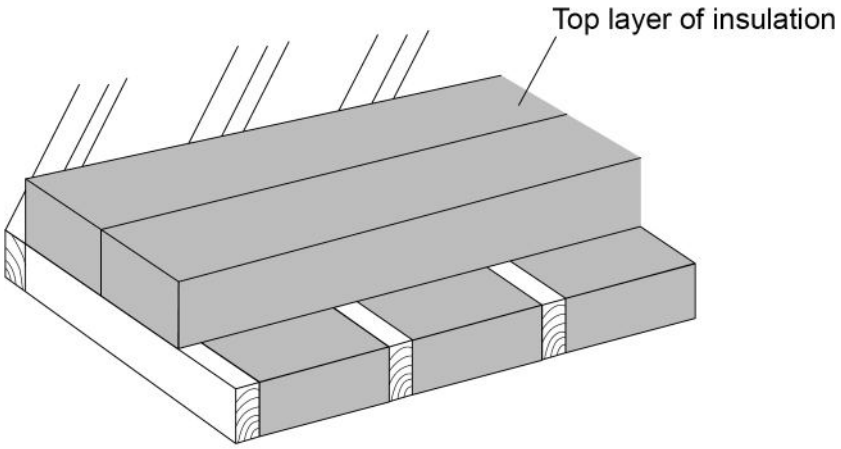
1 Put a bottom layer of insulation between the joists.



Subtract 10% from the area of the loft to allow for the joists.

Use the answer to work out how many rolls of bottom layer insulation you need.

2 Put a top layer of insulation over the joists and the bottom layer.

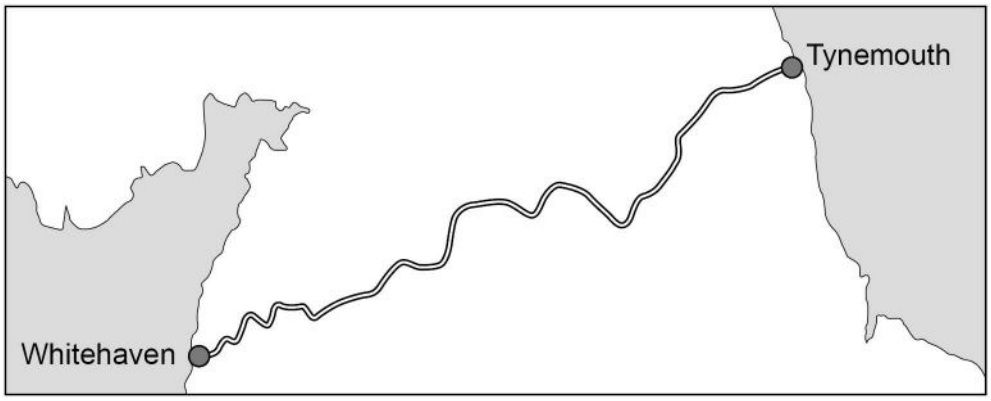


Use the **full** area of the loft to work out how many rolls of top layer insulation you need.

Data Sheet for Coast to Coast



The Coast to Coast cycle route starts at Whitehaven and finishes at Tynemouth.



The cycle route usually takes between 3 days and 5 days to complete.

Overnight stays

Here is some information about possible places for overnight stays along the route.

Place	Distance from Whitehaven
Whitehaven	0 km
Lorton	34 km
Keswick	50 km
Greystoke	77 km
Penrith	89 km
Melmerby	103 km
Alston	120 km
Allenheads	138 km
Rookhope	148 km
Stanhope	156 km
Consett	179 km
Newcastle	206 km
Tynemouth	225 km

Distance between places

The distance between places can be worked out from the table.

For example, the distance from Alston to Rookhope is 28 km

Baggage transfer

You can pay to have your baggage moved from one place to another.

END OF DATA

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Answer **all** questions in the spaces provided.

Do not write outside the box

1 Heating

There is a **data sheet** for Heating.

1 (a) Jack and Ann lived together in 2017
Jack was born in 1930
Ann was born in 1951

How much in **total** was their winter fuel payment?

Circle your answer.

[1 mark]

Q1 Worked Solution:
See official AQA mark scheme method.

£250 £300 £350 £500

[Method & Accuracy]

1 (b) Leon was born in 1951 and lives alone.

To pay his 2018 winter fuel bill he will

use his 2017 winter fuel payment

and

save some money each week for 12 weeks.

He expects his 2018 winter fuel bill to be £320

He says,

"If I save £10.50 each week I will have enough to pay a £320 bill."

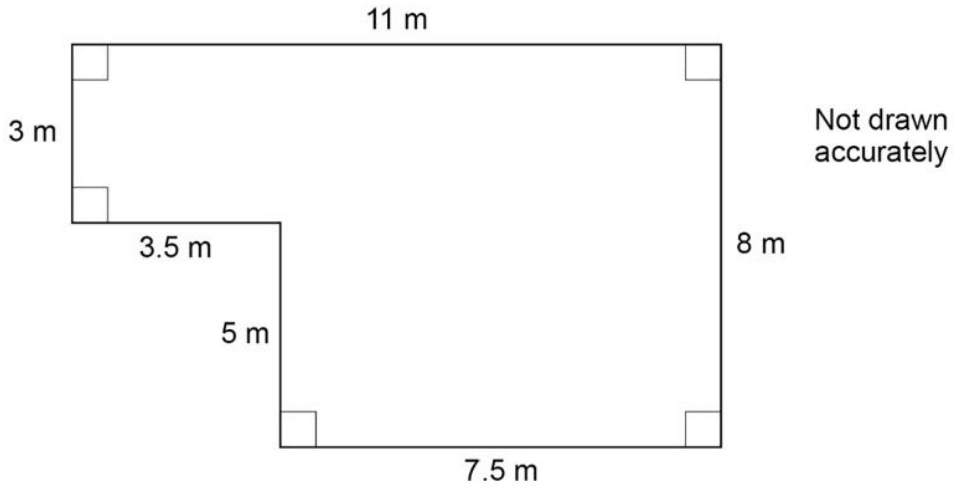
Is he correct?
You **must** show your working.

[4 marks]

Question 1 continues on the next page

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Prita's house has an L-shaped loft with no insulation.



1 (c) Show that the area of the loft is 70.5 square metres.

[2 marks]

Prita decides to insulate her loft.

She reads the instructions on the data sheet.

1 (d) Subtract 10% from the area of Prita's loft.

[2 marks]

Do not write outside the box

1 (e)

Bottom layer insulation

£26 per single roll

One roll covers an area of 11 square metres

Special offer

Buy more than 5 rolls and ***all*** *rolls are half price*

Top layer insulation

£24.75 per single roll

One roll covers an area of 6.5 square metres

Special offer

Pack of 4 rolls £80

Prita buys the packs and single rolls she needs in the cheapest way possible.

How much does she pay in total?

[7 marks]

16

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2 **Coast to Coast**

There is a **data sheet** for Coast to Coast.

Tim and Maisy live in London.

They are planning a holiday cycling the Coast to Coast route.



Maisy **Tim**

Tim makes these notes.

Monday	Travel by train from London to Whitehaven Overnight stay at Whitehaven
Tuesday	Start the Coast to Coast route Overnight stay
Wednesday	Cycle further along the route Overnight stay
Thursday	Cycle further along the route Overnight stay
Friday	Finish the route Overnight stay in Tynemouth
Saturday	Travel by train from Tynemouth to London

Use baggage transfer between each overnight stay

Do not write outside the box

2 (a) Here are the costs for their holiday.

Train from London to Whitehaven	£51.00 per person
Overnight stays	£35.00 per person per night
Other costs	£20.00 per person per day for 6 days
Train from Tynemouth to London	£54.50 per person
Baggage transfer	£8.50 per transfer

Tim and Maisy only have one piece of baggage between them.

Tim says,

"The total cost of our holiday will be **less than** £800"

Is he correct?
You **must** show your working.

[5 marks]

Question 2 continues on the next page

Do not write outside the box

Tim and Maisy will

cycle about the same distance on each of Tuesday, Wednesday and Thursday

stay overnight at Stanhope on Thursday night.

2 (b) Complete the table.

[3 marks]

Q2 Worked Solution:

See official AQA mark scheme method.

[Method & Accuracy]

Day	Start	Finish	Distance cycled that day (km)
Tuesday	Whitehaven		
Wednesday			
Thursday		Stanhope	

Do not write outside the box

2 (c) On Friday, Tim and Maisy will

cycle at an average speed of 30 km per hour
stop in Newcastle for 2 hours.

Maisy says,

"If we leave Stanhope at 11 am we will be in Tynemouth by 3.20 pm"

Is she correct?
You **must** show your working.

[5 marks]

13

Turn over for the next question

Do not write outside the box

3 Chocolate eggs



Carly

†3 (a) To make the eggs, Carly needs to heat chocolate to 43 degrees Celsius.
Her thermometer only measures in degrees Fahrenheit.

Use this formula to convert degrees Celsius to degrees Fahrenheit.

$F = 1.8C + 32$

F is the temperature in degrees Fahrenheit

C is the temperature in degrees Celsius

Q3 Worked Solution:

Convert 43 degrees Celsius to degrees Fahrenheit.

[2 marks]

2 × 54 or 36
allow 0.66... × 54 or 0.67 × 54
do not allow 0.6 × 54 or 0.7 × 54

their 36 × 3.5(0) +
(54 – their 36) × 2.5(0)
or their 36 × 3.5(0) + 18 × 2.5(0)

Check your answer.
Show how you have done your check.

[1 mark]

or 126 + 45 or 171 or 316.25
their 36 must be an integer < 54

their 104 + their 41.25
+ their 126 + their 45 – 150

[M1] [Rc] [M1] [Aa] [M1] [Aa]

Do not write outside the box

3 (b) One week, Carly sells these eggs.

	White chocolate	Milk chocolate	Dark chocolate
Small eggs	20	42	36
Large eggs	13	25	16

How many **more** milk chocolate eggs than dark chocolate eggs does she sell?
Circle your answer.

[1 mark]

9 15 17 44

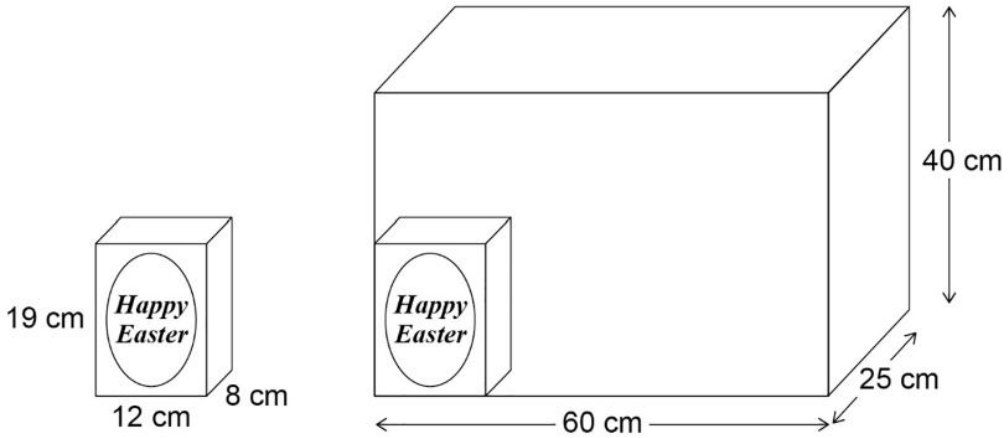
Question 3 continues on the next page

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3 (c) Carly puts each large egg in a box.

She packs the boxes into crates.

The boxes are all packed in the same way as shown.



Work out the **maximum** number of boxes she can fit in a crate.

[4 marks]

Do not write outside the box

3 (d) Carly takes 98 small eggs and 54 large eggs to an Easter fair.

She sells

65 small eggs for £1.60 each
the rest of the small eggs for £1.25 each

$\frac{2}{3}$ of the large eggs for £3.50 each

the rest of the large eggs for £2.50 each.

The total cost of making the eggs was £150

She says,

"My profit is **more than** £180"

Is she correct?
You **must** show your working.

[6 marks]

14

Turn over for the next question

Do not write outside the box

4 Competition



My students are taking part in a competition. I need a test paper for each student.

Mrs Scott

†4 (a) Mrs Scott needs 163 test papers.

The papers are in packs of 5

How many packs does she need?

[2 marks]

Q4 Worked Solution:

In particular, individual marks are mapped onto the following skills standards.
Representing Making sense of the situations and representing them.
A learner can:
Understand routine and non-routine problems in familiar and unfamiliar contexts and situations.

Check your answer.

Show how you have done your check.

[1 mark]

Identify the situation or problems and identify the mathematical methods needed to solve them.
Choose from a range of mathematics to find solutions.
Analysing
Processing and using the mathematics.
[Ra] [Rb] [Rc] [Aa] [Ab]

Do not write outside the box

4 (b) Kim and Ellie do some practice papers.

Here are their marks.

Kim	61	50	54	53	63	56	50	55
Ellie	51	54	62	57	60	55		

Ellie says,

"On average, I got higher marks than Kim."

Is she correct?
You **must** show your working.

[4 marks]

4 (c) Each question on the test paper has five answers to choose from.
For one question, Kim guesses the answer at random.

What is the probability that her guess is **not** correct?

[1 mark]

Do not write outside the box

4 (d) There are 15 questions on the test paper.

Here are the scoring instructions.

	Correct answer	**Incorrect answer**	**No attempt (–)**
Questions 1 to 5	5 points	0 points	0 points
Questions 6 to 10	6 points	–1 point	0 points
Questions 11 to 15	6 points	–2 points	0 points

The grid below shows the correct answers and Kim's answers.

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**13**	**14**	**15**
Correct answer	B	D	D	E	C	A	D	A	E	E	B	C	E	C	D
Kim's answer	B	D	A	E	C	B	D	A	A	–	E	–	–	C	C
Points															

Work out the **total** number of points Kim scores.

[4 marks]

Do not write outside the box

4 (e) Altogether, 84 000 students take part in the competition.

15% of the students win an award.

The awards are gold or silver.

There are four times as many silver awards as gold awards.

Is the number of **silver** awards **more than** 10 000?
You **must** show your working.

[5 marks]

END OF QUESTIONS

17

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ANSWER IN THE SPACES PROVIDED

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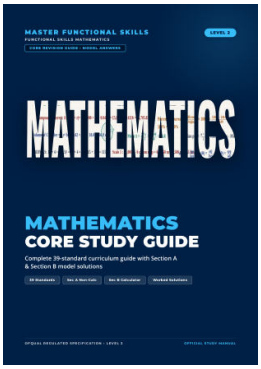
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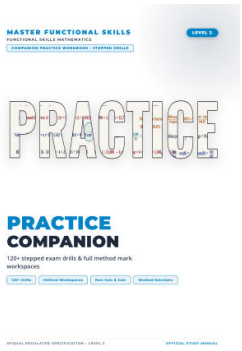


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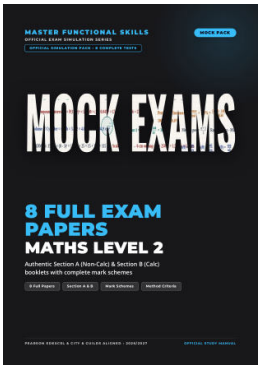


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